

400-Gb/s PAM4 Operation of an InP-on-Si Membrane EA Modulator Sub-Assembly with 1.0-V Differential Swing Voltage

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Abstract We demonstrate a 400-Gb/s PAM4 operation of a differential drive membrane electro-absorption modulator (EAM) sub-assembly with 1.0-V differential amplitude. 3-dB electro-optic bandwidth of >110 GHz is realized owing to a low capacitance 150- μ m-long EAM flip-chip-bonded ceramic substrate as well as an integrated 80- Ω differential termination. ©2026 The Author(s)

Introduction

The rapid growth of artificial intelligence and machine learning workloads is driving the demand of increasing bandwidth of optical interconnects in data centres as well as suppressing the link energy. Co-Packaged Optics (CPO) has emerged as promising packaging architecture to increase bandwidth density and improve energy efficiency [1]. Optical devices for CPO require a compact size, low power consumption, and a modulation capability of 200 Gb/s and beyond per lane [2,3]. Electro-absorption modulators (EAMs) and EA-DFB lasers have been regarded as attractive candidates from their advantages of high speed, a small footprint, and low power consumption. So far, the modulation capability of single-ended EAMs and EA-DFB lasers has been reaching 200 Gb/s and beyond per lane [4,5].

In recent research, InP-on-Si membrane EAMs and EA-DFB lasers with high optical confinement and low capacitance have achieved 3-dB bandwidth of 103 GHz and a 448-Gb/s PAM4 operation without termination resistors [6]. To further reduce the power consumption and improve the signal integrity, a differential voltage drive has been expected, as well as suppressing electrical crosstalk [8–10]. Most differential drive EA-DFB lasers employ either a single-EAM anode–cathode structure or a dual (tandem) EAM structure. The tandem structure enables high-speed operation, as demonstrated by 360-Gb/s PAM4 operation [10], although it requires a relatively complex device structure. In contrast, the single-EAM anode–cathode structure can be realized using a device structure similar to that of conventional EAMs, offering simpler implementation. However, the effective load resistance approximately doubles, which poses a challenge for achieving high-speed response.

In this paper, we report an InP-on-Si membrane EAM employing an anode–cathode differential driving structure. The membrane EAM utilizes a thin lateral p–i–n heterostructure, which

realizes low capacitance and a symmetric structure suitable for differential operation. Although the membrane EAM has the advantage of inherently low capacitance, a differential drive membrane EAM has increased an RC time constant compared to the single-ended EAM structure, which results in degradation of modulation bandwidth. To achieve a 400 Gb/s operation using the differential drive EAM structure, differential impedance matching of the EAM is required to suppress the bandwidth degradation. The sub-assembly demonstrates 3-dB bandwidth of >110-GHz and 400-Gb/s PAM4 operation with a low differential voltage of 1.0 V.

Design of differential drive membrane EAM sub-assembly

Fig.1 shows a schematic cross-sectional illustration of a differential drive membrane EAM sub-assembly. The membrane EAM is flip-chip-bonded on a ceramic (Al_2O_3) substrate, where electrodes of the membrane EAM and the substrate were connected using Au bumps, to achieve high-density integration and keep signal integrity. The membrane EAM has a thin InP layer with a thickness of approximately 230 nm, which is directly bonded onto a SiO_2 /Si substrate. A 400-nm-wide InGaAlAs core layer is buried in the thin InP membrane. P-type and n-type InP layers are formed on the left and right sides of the core layer, allowing a voltage to be applied from both sides. This structure realizes low capacitance and a large optical-confinement factor.

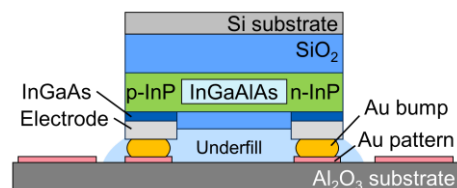


Fig. 1: Schematic cross-sectional illustration of a differential drive membrane EAM sub-assembly.

Light confined in the core is attenuated by applying a reverse bias voltage to the electrodes of the EAM, inducing absorption through the Franz–Keldysh effect. A short modulator length of 150 μm sufficiently realizes a high modulation efficiency owing to its high optical confinement [7]. On the top surface of the Al_2O_3 substrate, a ground-signal-signal-ground (GSSG) coplanar-waveguide (CPW) line is formed, while no metal pattern is formed on the backside of the substrate. To ensure the mechanical robustness of the sub-assembly, an underfill material is filled between the membrane EAM and the Al_2O_3 substrate. Fig. 2 shows the schematic diagram of the differential drive membrane EAM sub-assembly. The anode and cathode electrodes of the membrane EAM are electrically connected to the two signal lines of the CPW. Each electrode is connected in parallel with a 40 Ω termination resistor. Differential signals are applied to the anode and cathode. Fig. 3 shows a photograph of the fabricated differential drive membrane EAM sub-assembly from a top-side view. To maintain the CPW propagation mode, an air cavity is formed by recessing the Kovar base under the CPW area of the sub-assembly.

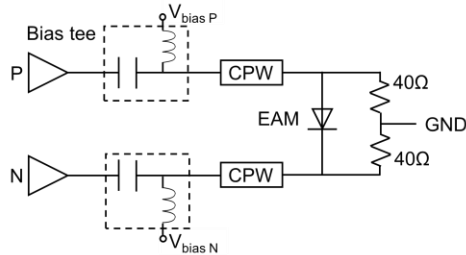


Fig. 2: Schematic diagram of a differential drive membrane EAM sub-assembly.

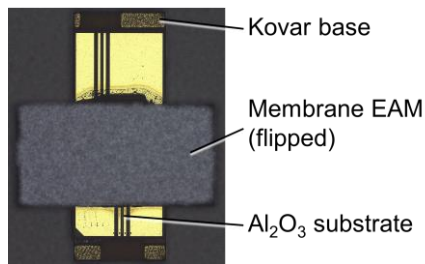


Fig. 3: Photograph of a differential drive membrane EAM sub-assembly from top-side view.

Measurement results

Fig. 4 shows DC characteristics of the membrane EAM sub-assembly at 25 $^{\circ}\text{C}$. Fig. 4 (a) shows a normalized output power characteristic as a function of bias voltage at a wavelength of 1305 nm. An extinction ratio (ER) of 3.72 dB was obtained within a voltage amplitude range between 0 V and 1 V, which is sufficiently higher than our targeted value of 3 dB. A high modulation efficiency of 3.7 dB/V is realized owing to optimised wavelength detuning [6]. Fig. 4 (b) shows an optical

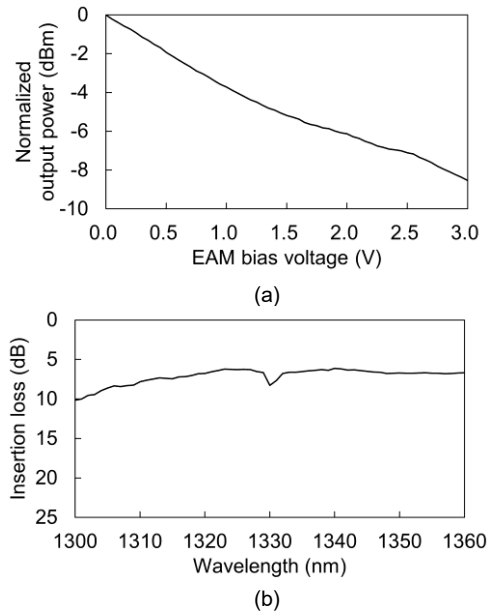


Fig. 4: DC characteristics of the membrane EAM sub-assembly. (a) Normalized output power. (b) Insertion loss.

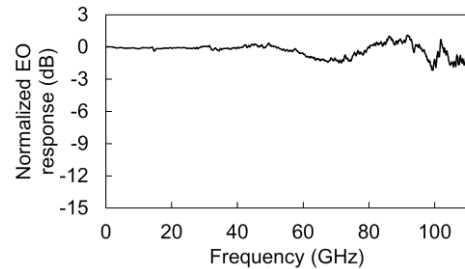


Fig. 5: Normalized EO response of the differential drive membrane EAM sub-assembly.

insertion loss (OIL) characteristic of the sub-assembly. We subtracted fibre-to-fibre coupling loss from the characteristic. The OIL characteristic is consistently <10 dB from 1300 nm to 1360 nm. Fig. 5 shows a measured EO response of the membrane EAM sub-assembly. The electrical signal from a vector network analyser (VNA) was applied to the GSSG transmission line on the substrate via a coaxial cable and an RF probe. 3-dB bandwidth of the membrane EAM sub-assembly was as wide as >110 GHz, thanks to the low device capacitance and reduced parasitic inductance enabled by the flip-chip bonding structure. Good impedance matching by using the termination resistors also contributed to increasing the bandwidth.

Fig. 6 shows a block diagram to measure optical eye diagrams. An arbitrary waveform generator (AWG) generates both positive and negative electrical signals to be used as a differential signal. The AWG was connected to two bias tees and an RF probe and two RF amplifiers through two RF cables. A reverse bias voltage of 0.5 V was applied between the anode and cathode via the bias tees. The frequency responses of the RF

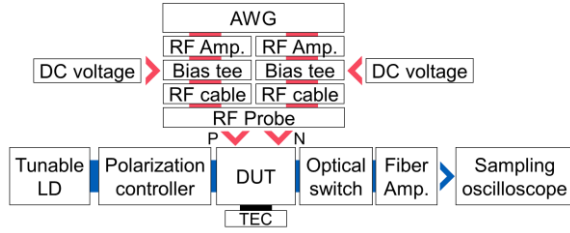


Fig. 6: Block diagram to measure optical eye diagrams.

amplifiers, bias-tees, RF cables, and the RF probe were calibrated using an embedded function of the AWG. The EAM was operated with a $2^{11}-1$ pseudo-random binary sequence quaternary signal at 25 °C. Electrical and optical eye diagrams were measured by a sampling oscilloscope and filtered with a half-baud-rate 4th-order Bessel–Thomson (BT) filter. The electrical eye diagrams were then equalized by a 15-tap feed-forward equalizer (FFE). The optical eye diagrams were equalized by the same 15-tap FFE. We also adopted a combined equalizer consisting of a 15-tap FFE and a 1-tap decision-feedback equalizer (DFE) in reference to the IEEE 802.3dj draft [11]. A target symbol error rate for the transmitter and dispersion eye closure quaternary (TDECQ) was set to 9.6×10^{-3} , which is compliant with the IEEE802.3dj draft as well. Fig. 7 shows the 400-Gb/s PAM4 differential electrical eye diagram output from the AWG. We observed clear eye opening and a differential voltage of 1.0 V. Fig. 8 shows the optical eye diagrams in back-to-back. Even operated with a low differential voltage of 1.0 V, the 400-Gb/s PAM4 eye diagram was opened as shown in Fig. 8 (a). We also obtained a high ratio level mismatch (RLM) of 0.93, which is close to the ideal value of 1.0. Fig. 8 (b) shows the optical eye diagram when an additional 1-tap DFE was applied. With the stronger equalization enabled by DFE, the eye opening is further improved compared with the FFE-only case while maintaining a high RLM of 0.93. Fig. 9 shows TDECQ characteristics as a function of operating bitrate for both optical and electrical eye diagrams in back-to-back. Open squares show TDECQ values of the optical eye diagrams equalized with a 15-tap FFE, and crosses show those of the optical eye diagrams equalized with a 15-tap FFE and an additional 1-tap DFE. As a reference, open circles show those of the electrical eye diagrams equalized with a 15-tap FFE for the measurement setup. TDECQ values increase as increasing a bit rate for all conditions. When adopting the combined equalizer (FFE + DFE), TDECQ values remain below 3.4 dB up to 340-Gb/s PAM4 operations, which inherently includes the characteristics of the measurement setup (i.e., a TDECQ value of 1.8 dB). When operated with a 400-Gb/s PAM4 signal, a TDECQ value was

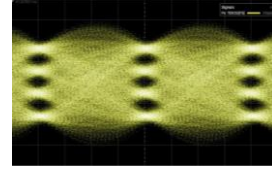


Fig. 7: Differential electrical eye diagrams for 400Gb/s PAM4.

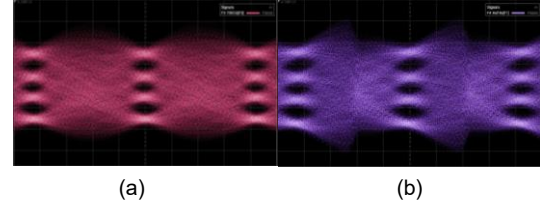


Fig. 8: Optical eye diagrams in back-to-back. (a) with 4th-order BT filter and 15-tap FFE, (b) with 4th-order BT filter, 15-tap FFE and 1-tap DFE.

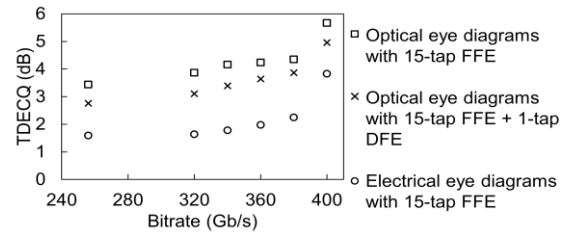


Fig. 9: TDECQ characteristics as a function of operating bitrate.

increased to 5.7 dB for the optical eye diagram with the 15-tap FFE, representing a rapid increase from 4.3 dB at a 380-Gb/s PAM4 operation. When adopting the 1-tap DFE, the TDECQ value decreased to 5.0 dB. The TDECQ value of the electrical eye diagram also increases sharply from 2.3 dB to 3.8 dB at a 400-Gb/s PAM4 operation. It indicates that 400-Gb/s PAM4 optical eye diagrams are strongly influenced by the measurement setup.

Conclusion

We demonstrated a 400-Gb/s PAM4 operation of a differential drive InP-on-Si membrane EAM sub-assembly using 1.0-V differential amplitude. 3-dB EO bandwidth of >110-GHz was realized owing to a low-capacitance 150-μm-long EAM flip-chip bonded ceramic substrate as well as an integrated 80-Ω differential termination. When applying a low differential voltage of 1.0 V, we observed clear 400-Gb/s PAM4 optical eye opening as well as TDECQ values of 5.7 dB (15-tap FFE) and 5.0 dB (15-tap FFE + 1-tap DFE) in back-to-back. A TDECQ for the measurement setup was as high as 3.8 dB at 400-Gb/s PAM4. Its influence is reflected in the TDECQ values of the optical eye diagrams. Therefore, optical eye diagrams may exhibit lower TDECQ values practically. These results show that this differential drive membrane EAM subassembly will be suitable for a CPO transceiver which exhibits energy-efficient, high-density and 400-Gb/s/lane.

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